



Krishna Kumar Sundararaj

Phone number: (+91) 9361849056 (Mobile) | **Email address:** krishnasundararaj97@gmail.com | **LinkedIn:**

<https://www.linkedin.com/in/krishnasundararaj/> | **Address:** Bengaluru, India (Home)

● ABOUT ME

Clinical research professional with hands-on experience supporting multi-centre ICU-based interventional and medical device trials. My work has involved close coordination with multidisciplinary teams, managing clinical data, and ensuring studies run smoothly in high-acuity settings. With an MSc in Cancer Medicine from Queen's University Belfast, I bring a foundation in neurocritical care research and translational oncology, along with practical experience in ICH-GCP-compliant trial coordination, regulatory documentation, and patient recruitment.

● WORK EXPERIENCE

PROJECT RESEARCH SCIENTIST I – ST. JOHN'S RESEARCH INSTITUTE – 02/05/2025 – 31/03/2026 – BENGALURU, INDIA

1. ICMR-Funded Multi-Centre ICP/ CPP Trial (Prospective Interventional Study)

- Serving as Research Scientist for a two-centre clinical trial (Total N=206; Site Target N=83)
- Screened 40+ severe TBI patients; enrolled 11 participants, independently conducting informed consent in ICU settings
- Operating ICM+ cerebral physiological monitoring software for real-time data acquisition
- Managing CRF completion, source documentation, and data quality verification
- Identifying and reporting AEs/SAEs per ICH-GCP guidelines, preparing and submitting IEC continuing reviews, amendments, and progress reports
- Maintaining an audit-ready Investigator Site File (ISF), coordinating intensivists, neurosurgeons, statisticians, and research nurses

2. Prospective Medical Device Studies (Completed; N=30 each)

- Coordinated two ICU-based device evaluation studies from initiation to study close-out
- Managed 60 total participants across both studies
- Led regulatory submissions, participant recruitment, compliance oversight, and final reporting
- Ensured ISO 14155-aligned documentation and GCP compliance

3. Retrospective Polytrauma Outcomes Study (Target N=140)

- Leading structured data abstraction and database development
- Coordinating cross-departmental chart review and data validation
- Supporting statistical analysis, planning and manuscript preparation
- Targeted completion by March 2026

4. International Quality Improvement Collaborative (Stanford–NCG EQulP India)

- Conducted a structured qualitative assessment of ICU referral pathways
- Contributing to the implementation of system-level palliative care integration
- Supporting measurable improvement initiatives within neurocritical care

IMMUNOASSAY DEVELOPMENT SCIENTIST – RANDOX LABORATORIES – 08/05/2023 – 31/07/2024 – ANTRIM, UNITED KINGDOM

- Optimized nanoparticle-based immunoassays, achieving 15% reduction in variance.
- Performed buffer preparation, filtration, conjugation, and purification aligned with GLP and ISO 13485 standards.
- Maintained full traceability and documentation using Q-Pulse software.
- Supported internal audits and contributed to regulatory compliance readiness.

CLINICAL RESEARCH ASSISTANT – HINDUSTHAN HOSPITAL – 20/08/2020 – 30/04/2021 – COIMBATORE, INDIA

- Screened and enrolled participants for COVID - 19 vaccine clinical drug trial collaborate with Dr Reddy's Laboratory.
- Improved patient recruitment rates by 30% through enhanced coordination.
- Documented data in Oracle Clinical and ensured protocol adherence and ethical compliance.



ANAESTHESIA TECHNOLOGIST – GANGA ORTHOPAEDIC HOSPITAL – 04/02/2018 – 31/10/2018 – COIMBATORE, INDIA

- Assisted in complex orthopedic surgeries, ensured correct anesthesia dosing, and machine prep.
- Improved OR readiness by standardizing pre-op safety checklists.

ANAESTHESIA TECHNOLOGIST – APOLLO CANCER INSTITUTE – 10/11/2018 – 03/08/2019 – CHENNAI, INDIA

- Assisted in over 150 oncology surgeries, maintaining <1% adverse event rate.
- Managed OT readiness and post-operative ICU transfers.
- Supported anesthesia administration and equipment handling in high-risk cases.

PROJECTS

02/05/2025 – 31/03/2026

OPT-STAND-CPP: Optimising Cerebral Perfusion Pressure in TBI (ICMR)

11/07/2025 – 31/03/2026

Impact of Surgery Timing in TBI with Polytrauma (Retrospective study)

15/06/2025 – 31/03/2026

EqUP India 2025 – Quality Improvement Collaborative (Stanford–NCG)

03/06/2025 – 15/10/2025

CytaCoat Foley Catheter Trial – Infection Control Innovation

09/06/2025 – 11/10/2025

UbiqVue Wireless Monitoring Trial – Device Comparison Study

08/05/2023 – 31/07/2024

Nanoparticle-Based Immunoassay Optimization – Diagnostic R&D Project

08/10/2021 – 31/07/2022

Prostate Cancer Nanoparticle Implants – MSc Research Project

PUBLICATIONS

In preparation/under review

1. A Scoping Review Procedure for Using Cerebral Perfusion Pressure to Predict Disability Outcomes Following Traumatic Brain Injury
2. Early Palliative Care Referrals to Catastrophic Neurological Conditions in the ICU: A Scoping Review
3. Implementing Optimised Cerebral Perfusion Pressure Targets in Traumatic Brain Injury: Early Clinical Outcomes and Workflow Feasibility in a Neurocritical Care Unit

SKILLS

Clinical Trial & Research Operations

Multi-centre RCT Coordination | ICH-GCP (E6 R2) & ISO 14155 Compliance | Informed Consent (ICU & Legally Authorized Representative) | AE/SAE Identification & Safety Reporting | Patient Screening, Recruitment & Follow-Up | CRF Management & Query Resolution | A3 Methodology | Scientific Literature Review & Manuscript Development | Quality Improvement (PDSA cycles, workflow redesign) | IRB/IEC Submissions & Regulatory Documentation | Study Start-Up, Continuing Review & Close-Out | Protocol Implementation & Monitoring Readiness

Clinical & Translational Expertise

ICU-Based Clinical Data Collection | Neurocritical Care Research (TBI, ICP/ CPP Monitoring) | Surgical ICU & Palliative Care Integration | Perioperative & Anaesthesia Clinical Experience | Medical Device Clinical Evaluation (Wearables, Catheters) | Translational Oncology Research (Nanoparticle Therapeutics) | Clinical Outcomes Assessment & Workflow Feasibility Studies



Data, Regulatory & Laboratory Systems

ICM+ Cerebral Monitoring Platform | GraphPad Prism (Biostatistics & Data Visualization) | Oracle Clinical (EDC) | 3d Bioprinting | Q-Pulse Quality Management System | Microsoft Office Suite & Access | Zotero (Reference Management) | Buffer Preparation | Diagnostic Product Pipeline Support for Global Markets | ISO 13485 Quality Management Systems | Tangential Flow Filtration | Drug Delivery System Design & Translational Oncology Applications | Polycaprolactone (PCL) Biodegradable Implant Fabrication | Physicochemical Characterization of Biomaterials | In Vitro Biocompatibility & Cytotoxicity Testing | Cell Culture & Biocompatibility Testing | Experimental Design & Controlled Laboratory Studies | Laboratory GLP Documentation & Traceability | Batch Traceability & Regulatory Audit Preparation | Nanoparticle Conjugation & Assay Optimization

EDUCATION AND TRAINING

15/09/2021 – 15/09/2022 Belfast, United Kingdom

M.SC. (RESEARCH) IN CANCER MEDICINE Queen's University Belfast

Website www.qub.ac.uk | Level in EQF EQF level 7

16/07/2014 – 10/10/2017 Chennai, India

B.SC. IN OPERATION THEATRE AND ANAESTHESIA TECHNOLOGY Dr. MGR Medical University

Website www.tnmgrmu.ac.in | Level in EQF EQF level 6

CONFERENCES AND SEMINARS

07/12/2025 – 07/12/2025 St. John's Medical College Hospital, India

TRACE 2025 - Trauma Analgesia & Critical Care Education

02/11/2025 – 02/11/2025 GKNM Hospital, India

3C Conclave 2025

13/09/2025 – 13/11/2025 St. John's Medical College Hospital, India

Primary and Secondary Medical Board Activation Workshop in Palliative Care

11/06/2025 – 11/06/2025 St. John's Medical College Hospital, India

Ethics in Palliative Care Workshop

NETWORKS AND MEMBERSHIPS

18/09/2025 – CURRENT India

Society of Neuro Critical care

22/07/2025 – CURRENT Brussels, Belgium (EU-wide)

Member – COST Action CA22114 (TREASURE Network)

03/08/2022 – CURRENT India

Lifetime Member – Association of Allied Health Professionals (AAHP)

CERTIFICATIONS

Certifications Till date

- Introduction to the Principles and Practice of Clinical Research - National Institutes of Health, Sep 2025
- Fire and Safety - St John's Medical college hospital - Jul 2025
- Data management for clinical research studies - The Global Health Network (Jul 2025 – Jul 2027)
- Lean Six Sigma White Belt – Aveta Business Institute, Jul 2025
- Digital Health Foundation Course – National Cancer Grid India, Jul 2025
- Research to Publication – UCSF Health, Jun 2025
- ICH-GCP (E6 R2) – The Global Health Network (May 2025 – May 2027)
- IELTS - 6.5 (Overall) - IDP Education consultancy, Feb 2025
- Certified Anaesthesia Technologist – AAHP, Aug 2022
- BLS, AED & Manual Handling – Safe2Care Training, Jan 2022